
CERTIFICATE OF PARTICIPATION

DATA SCIENCE HACKATHON

March 14, 2026

Participant: Jakob Bell

Jakob demonstrated exceptional technical aptitude and problem-solving skills while managing a massive dataset comprising nearly one billion rows of AIS data spanning several years, sourced from the Danish Maritime Authority.

Handling geospatial data at this scale presents unique challenges in terms of memory management and processing speed. The participant addressed these by:

- ❖ **Optimizing Workflows with Polars:** Leveraging the Polars library to utilize its lazy evaluation and multi-threaded query engine, ensuring that data transformations remained performant despite the billion-row scale.
- ❖ **Strategic Data Architecting:** Implementing efficient data partitioning and filtering techniques to navigate years of maritime records without bottlenecking system resources.
- ❖ **Analytical Rigor:** Moving beyond simple data processing to extract meaningful insights from complex maritime trajectories and signal patterns.

Beyond the technical challenges, Jakob consistently demonstrated exceptional teamwork and adaptable problem-solving, effectively communicating complex technical challenges and solutions to both technical and non-technical matters. Jakob played a key role in:

- ❖ **Visualizing Routes of Ships:** Exploring methods for the clear and intuitive visual representation of complex maritime trajectories.
- ❖ **Creating Interactive Dashboards:** Committing to the development of interactive dashboards to democratize access to the insights derived from the AIS data.
- ❖ **Investigating Suspicious Fishing Behavior:** Proactively applying analytical rigor to flag and investigate suspicious fishing activities in prohibited areas.
- ❖ **Addressing Missing Data:** Showing persistence and methodological skill in developing strategies to identify and impute missing data points, thus maintaining the integrity and completeness of the billion-row dataset.

The participant's contributions were meaningful and significantly advanced the project goals.



Sergey Belomestnykh

Chairperson Social Data Science Subject Council
+49 152 04139429
qmn283@alumni.ku.dk